



北京航空航天大学

科学写作与报告

引言与文献综述

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本课程的基本内容：

- Review：标题、摘要与关键词
- 科技论文的写作技巧（中）

Review: 标题、摘要与关键词

- 标题的基本要求：准确、简洁、清楚、易读
 - 摘要的结构：IMRAD（背景、目的、方法、结果和结论）
 - 关键词的确定方法：基于论文的核心概念和研究方法
 - 作者署名、排序及单位：贡献度
 - 其他相关信息：作者信息、基金支持、审稿周期
-

科技论文的写作技巧（中）

- 1. 引言 Introduction
- 2. 文献综述

1. 引言 Introduction

- 1.1 基本概念
- 1.2 写作思路
- 1.3 写作技巧
- 1.4 评估

1.1 Introduction基本概念

- 主要叙述论文的研究目的和研究背景，使读者对论文研究问题的进展和概况以及论文的内容有基本了解。
 - 研究背景（**综述**）：对相关研究领域的历史、现状和进展进行一定的综述和总结，指出尚未解决的问题和不足，引出研究目的。
 - 研究目的与方法：说明研究中所要解决的问题和所采取的手段。
 - 研究贡献：列出论文对相关领域研究的贡献。
 - 论文结构：简要说明论文篇幅的结构。

1.1 Introduction基本概念：作用

- 讲述论文研究的背景知识，使读者无须阅读其他文献也能够很好地理解和评估该论文的结果。
- 仔细选择参考文献以梳理论文的研究思路，介绍研究的重要性，简单、清晰地阐述论文的研究目的。
- 阐述本领域前沿进展及未解决的难题，相关的研究工作及其结论，以及本文如何针对现有问题开展研究。

1.1 Introduction基本概念： 相关名词

■ 研究领域：

- 指科学论文以及更广泛的同行所研究的更大的科学方向。
- 如： **Information Theoretic Security**

■ 研究方向：

- 指科学论文以及其他同行针对同一科学问题所研究的泛化的主题。
- 如： **Covert Communications**

■ 研究主题：

- 指科学论文所研究的科学内容。
- 如： **Second-Order Asymptotics of Covert Communications over AWGN Channels**

1.1 Introduction基本概念：结构

■ 科学背景：

- 阐述该研究领域的科学意义，是否有重要科学问题亟须解决，能否推动人类的认知，是否有广阔的应用前景等。

■ 同行研究：

- 评述同行研究人员针对该主题的重要研究。

■ 待解决问题：

- 批判性地分析本研究方向，是否有未解决的问题，是否有待改进或攻关的难题，是否有应用的前景，是否需要改进研究方法等。

■ 本研究的方案：

- 如何研究新方法、新思路、新应用。

1.2 Introduction写作思路

- 科学背景：确定研究主题
 - 将研究领域内更加宽泛的重要科学问题作为科学背景，阐述研究领域的科学意义。
 - 无须深度介绍研究的细节，也无须做批判性评论，只需使用陈述性语言，清晰地阐述科学领域和研究意义，凸显重大科学问题和科学难题，合理地借助参考文献，构建科学思路和脉络。

1.2 Introduction写作思路

■ 同行研究：批判性文献综述

- 从科学背景和研究领域专注到研究方向，需要分析在该研究方向内的重要贡献。
- 研究领域内同行的重要研究须按一定的逻辑择要评述，不需要按照时间轴整理，根据研究内容的不同侧重点展开论述。
- 适当提及反对方，让文章更加客观，也彰显作者的学术包容性和公正性。

1.2 Introduction写作思路

- 待解决问题：强调科学原理
 - 将讨论范围聚焦于研究主题。通过对同行研究的评述，发掘同行研究未考虑全面的问题、暂未分析过的内容，或者研究中的不足之处。针对待解决的科学问题提出自己的解决方案或者思路。
 - 严格遵循科学原理。评判其他研究的不足或未涉及的部分，展示预期研究内容的科学思路的正确性与合理性。

1.2 Introduction 写作思路

- 本研究的方案：如何研究待解决的问题
 - 根据前面的科学原理，针对待解决的问题设计合理的研究方案。
 - 不必深度描述研究方案的细节，只需讲明研究方案的科学性、合理性和优越性。

1.2 Introduction写作思路

- 论文结构简介：介绍论文的章节结构
 - 引言的结尾部分可以强调本文重要的结论。
 - 最后用小段简短的文字（3 - 4句话）介绍论文的结构。
 - 如：本文第2节详细介绍了系统模型，第3节总结了本文的主要结果，第4节分析了结论。

1、科学背景，确定研究主题及意义

Iterative NOMA Detection for Multiple Access in Satellite High-Mobility Communications

Jiawei Wang[✉], Graduate Student Member, IEEE, Chunxiao Jiang[✉], Senior Member, IEEE, and Linling Kuang[✉], Member, IEEE

Abstract—Non-orthogonal multiple access (NOMA) is a promising technology for next generation multiple access (NGMA). However, traditional NOMA detection methods cannot cope with challenges of NGMA in satellite high-mobility communications. On the one hand, due to the high mobility and heterogeneity of terminals in terms of the velocity and acceleration, time-varying Doppler shifts caused by high-mobility terminals are relatively higher and different for each user, which degrades the demodulation performance seriously and makes multi-user interference cancellation become the bottleneck. On the other hand, owing to the wide distribution of high-mobility terminals, the arriving time of user signals is different at the receiver, which further incurs more difficulties for NOMA detection. To solve such a problem in satellite high-mobility communications, we propose a novel multi-user detection method based on the new three-dimensional (3D) factor graph for high-mobility environments, named the turbo iterative detection (TID) algorithm. Specially, the proposed algorithm consists of the interference cancellation loop, the Doppler elimination loop and the decoding loop. By means of message passing along edges in the proposed 3D factor graph, these three iterative loops can interact with each other to effectively eliminate time-varying Doppler shifts of heterogeneous high-mobility terminals and interference among multiple users. Simulation results show that the proposed algorithm improves the bit error ratio (BER) performance more than 0.9 dB with less computational complexity compared with traditional algorithms, which demonstrates the superiority of this algorithm in terms of the BER performance and computational complexity.

Index Terms—NOMA, multi-user interference cancellation, satellite communications, factor graph, message passing.

I. INTRODUCTION

SATELLITE communications can provide various services in global coverage with its wide coverage ability [1], which has become imperative to move forward for future integrated satellite-terrestrial networks [2], [3]. With the increasing

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2、分析待解决的问题，强调背景的必要性

3、文献综述： 分析现状及不足

and power-domain (PD) NOMA. For the former, CD-NOMA is capable of obtaining gain due to the use of spreading sequences of codewords [19]. In addition, the transmitted bandwidth can be spread by spreading sequences of codewords to improve the anti-interference ability of the transmitted signal. While for the latter, it utilizes superposition coding at the transmitter and successive interference cancellation at the receiver to support multiple access [20]. However, most traditional NOMA detection algorithms cannot cope with NGMA challenges in satellite high-mobility communications. In [21] and [22], precoding and power allocation strategies were presented to improve PD-NOMA performance without considering mobility and asynchronous characteristics of multi-user signals. Then, the authors proposed user dissociation and re-association control methods to deal with mobility issues of NOMA [23]. Later, in [24], considering heterogeneous mobility profiles of different users, the authors proposed the orthogonal time frequency space (OTFS) based PD-NOMA system, where receiving strategies were also designed to remove interference and harvest both multi-path and multi-user diversity. Whereas, this approach could not support asynchronous multiple access and the receiver in the satellite was difficult to obtain diversity in the light of sight (LoS) channel. In addition, most of these works assumed perfect synchronization at the receiver [25]–[27], i.e., the arriving time of each high-mobility terminals' signal was guaranteed to coincide. These methods reduced data transmission efficiency seriously and required accurate cooperation of each high-mobility terminal, which could not be achieved in satellite high-mobility communications. In the asynchronous case, the authors proposed a pre-coding and whitening-and-decomposing (WD) detection scheme [28]. However, this scheme assumed that the whole information of each terminal was known to the proposed precoding operator. Considering wide geographical distribution of high-mobility terminals, this assumption was impractical in high-mobility communication environments. Later, in [29], the authors proposed an asynchronous transmission system for multiple access channels, using the trellis-based detection methods to improve the performance. Nevertheless, the high Doppler frequency offset and the Doppler rate were not considered in this algorithm, therefore this scheme could not be applied in satellite high-mobility communication systems. Similarly, in [30], the authors proposed a novel triangular successive interference cancellation (T-SIC) that exploits a triangular pattern to perform the interference cancellation from all interfering users for the desired user without considering time-varying Doppler shifts in high-mobility environments. Different from aforementioned methods, in [31], jointly with asynchronous analog network coding (AANC), time offsets were added in the transmitter artificially to improve the performance of the cooperative communication system. Although these algorithms could exploit diversity gains to improve the system performance, high-mobility terminals would degrade diversity gains seriously, which inevitably leads to attenuation of the BER performance. Overall, these aforementioned multiple access approaches designed mainly for motionless or low speed ground terminals are not appropriate for NGMA in satellite high-mobility communications.

4、结合综述引出 本文要研究的问题

From above analysis, traditional multiple access detection methods have not taken both asynchronous arriving signals and time-varying Doppler shifts into consideration, which makes these methods difficult to be applied in satellite high-mobility communication systems for NGMA. Therefore, to cope with NGMA challenges mentioned above in satellite high-mobility communications, it is rather significant to develop the new asynchronous NOMA interference cancellation algorithm designed especially for satellite high-mobility communication systems, which is capable of achieving the better BER performance in high-mobility environments. Motivated by this, in this paper, we design the new factor graph considering the influence of high-mobility nature to cope with challenges of multiple access in high-mobility environments. Then a novel turbo iterative asynchronous NOMA detection algorithm is proposed to improve the interference cancellation performance for NGMA in satellite high-mobility communications. Concretely, the contributions of this paper are summarized as follows:

- We design a three-dimensional (3D) factor graph framework for asynchronous NOMA detection in satellite high-mobility communications, where different high Doppler frequency offsets and Doppler rates of heterogeneous high-mobility terminals are considered in the received signal. Then, based on the proposed factor graph model, we formulate passing messages of function nodes and variable nodes along edges in the proposed factor graph according to the sum-product rule.
- We propose the novel turbo iterative NOMA detection algorithm that consists of the Doppler elimination loop, the interference cancellation loop and the decoding loop. With the aid of the information interaction of these designed iterative loops in the proposed factor graph, the negative influence of time-varying Doppler shifts and multi-user interference can be eliminated effectively. In addition, the corresponding signal to interference plus noise ratio (SINR) analysis in high-mobility environments is also given.
- We evaluate the performance of the proposed algorithm by numerical simulations, which demonstrates that the turbo iterative NOMA detection algorithm can achieve the better BER performance with less computational complexity compared with traditional detection algorithms.

The rest of the paper is organized as follows. We first introduce the system model in Section II. Then, in Section III, we present the novel 3D factor graph in high-mobility environments. Later, in Section IV, passing messages of function nodes and variable nodes along edges in the proposed 3D factor graph are formulated. In Section V, based on the presented 3D factor graph, we propose the novel turbo iterative detection method of asynchronous NOMA for NGMA and give the SINR analysis of the proposed algorithm in high-mobility environments. Simulation results are then shown in Section VI. Finally, Section VII gives the conclusion.

II. SYSTEM MODEL

Fig. 1 presents the system model of asynchronous NOMA in satellite high-mobility communications. In the transmission

5、给出主要贡献（包括研究方案）

6、论文结构

1.3 Introduction写作技巧

- 引言中是否应该直接给出结论？
 - 欠缺经验的作者往往认为应该将重要发现写在论文最后的“结论”中，这是不太可取的写作方法。
 - 主要结果和结论的简述也是引言的核心部分，虽然跟摘要和结论的部分内容会重复，但可以让读者逐步印证论文的结论和观点。
 - 需要避免引言和摘要及结论中文字的直接重复，可采用不同的方式叙述。
 - 采用适当的方式强调作者在本研究中最重要发现和贡献，切忌像文学作品创作时故意制造悬念。

1.3 Introduction写作技巧

decoder side. More specifically, the major and novel contributions of our work are listed below.

- New achievability bounds (with both maximal probability of error and average probability of error) are obtained for cases with random coding and input constraints.
- New converse and achievability bounds are found on the channel coding rate over AWGN channels when the codewords are generated from Gaussian distribution and selected from a set with maximal power constraint.
- Normal approximations for both bounds over covert

分点阐述文章主要贡献，概述了主要结论

By analyzing the asymptotic behavior of our achievability and converse bounds, we show that under mild conditions on the fading distribution,²

$$R^*(n, \epsilon) = C_\epsilon + \mathcal{O}\left(\frac{\log n}{n}\right). \quad (4)$$

This results holds both for the case of perfect CSIT and for the case of no CSIT, and independently on whether CSIR is available at the receiver or not. By comparing (2) with (4), we observe that for the quasi-static fading case, the $1/\sqrt{n}$ rate penalty is absent. In other words, the ϵ -dispersion (see [9,

这篇文章偏向基础理论研究，作者在Intro部分详细解释了最主要结论，以便于读者跟上思路

1.3 Introduction写作技巧

- 背景介绍和问题提出时需要注意什么？
 - 尽量准确、清楚且简洁地指出所探讨问题的本质和范围。
 - 在背景介绍和问题的提出中，应引用“最相关”的文献以指引读者。
优先选择引用的文献包括相关研究中的经典、重要和最具有说服力的文献。力戒不恰当地大量引用作者本人的文献。
 - 针对不同读者群，考虑采取适当方式解释和定义专门术语或缩写词，以帮助编辑、审稿专家和读者阅读和理解。

1.3 Introduction写作技巧

I. INTRODUCTION

MULTIPLE-INPUT MULTIPLE-OUTPUT (MIMO) is one of the key technologies to achieve high-data rates in current wireless systems. Both the transmitter and the receiver are equipped with multiple antennas, which allows to spatially multiplex several streams of data and transmit them simultaneously. When the scattering environment is rich enough to allow receive antennas to separate out signals from different transmit antennas, MIMO channels offer large

开篇介绍主要研究点，
多输入多输出技术，强调其优势，便于后续在特定场景中引入

I. INTRODUCTION

CONSIDER a delay-constrained communication system operating over a slowly-varying fading channel. In such a scenario, it is plausible to assume that the duration of each of the transmitted codewords is smaller than the coherence time of the channel, so the random fading coefficients stay constant over the duration of each codeword [1, p. 2631], [2, Sec. 5.4.1]. We shall refer to this channel model as *quasi-static fading channel*.¹

When communicating over quasi-static fading channels at a given rate R , the realization of the random fading coef-

该研究的主要创新点在于理论场景的延伸拓展，因此开篇给出研究场景，并引入经典文献进行详细介绍

1.3 Introduction写作技巧

■ 如何进行批判性写作？

- 讲述该领域论文的一般研究思路，在以往研究中遇到的困难，以引出本研究如何解决该问题，以及如何设计研究思路。
- 写作时从多角度分析主题，评价同行的不同思路与观点，重要的是明确展示本研究的观点和立场。
- 叙述前人工作的欠缺以强调自己研究的创新时，应慎重而留有余地。应使用客观公正的语言评价他人工作，切忌使用有损他人情感和尊严的话语。
- 引言部分是全篇论文引用文献相对集中的部分，撰写时请特别留意正确的引证。

1.3 Introduction写作技巧

A number of works focused on improving the communication efficiency by various means, such as using channel uncertainty in [7][8][9], using jammers in [10][11] and other methods in [12][13]. These methods are discussed in the asymptotic regime. However, in practical communication, we are more concerned about the behaviors in finite blocklength regime. For example, given a finite block length n , how many information bits can be transmitted with a given covert criterion and maximal probability of error ϵ , under which the adversary is not able to determine whether or not the transmitter is communicating effectively. When the channels are discrete memoryless, this question has been answered by [14][15], where the exact second-order asymptotics of the maximal number of reliable and covert bits are characterized when the discrimination metrics are relative entropy, total variation distance (TVD) and missed detection probability with fixed probability of false alarm, respectively. For AWGN channels and slow fading channels, the maximal transmit power and the maximal transmit bits in finite block length are partly characterized in [16] and [17], respectively by applying the results in finite blocklength regime. However, it is inappropriate to directly apply the bounds in [18] for Gaussian signaling without proper discretion since the power constraint is not satisfied with probability 1 for Gaussian signalling in finite blocklength regime.

讲述该领域论文的一般研究思路，分析并总结已有论文的贡献

叙述前人工作的欠缺，引出本研究如何解决该问题，强调创新点

1.3 Introduction写作技巧

■ 与Abstract的区别

- 从内容目的上区分，摘要 是论文的缩影，它高度概括了论文的各个部分，重点是研究结果和结论。
- 引言 则是课题研究的必要说明，重点写选题的缘由、立题的依据和亟待解决的问题。
- 从结构位置看，摘要 是独立于论文之外的部分,是一篇完整的短文。
- 引言 是文章的一个组成部分,缺少引言,文章结构就残缺,该研究论文的展开就显得突兀、生硬,不能成为一篇构思严谨的科技论文。

1.4 Introduction评估

- 引言在整篇论文中所占的比例是否过高？
- 是否准确、清楚且简洁的指出问题的本质和范围？
- 背景信息是否都与论文目的有关？
- 是否清楚正在解决或试图解决什么问题，以及为什么选择特定方法？
- 是否与论文的主要结果和结论密切相关？
- 是否与摘要有足够的区分？

2. 文献综述:

- 2.1 基本概念
- 2.2 写作思路
- 2.2 写作技巧
- 2.3 评估

2.1 Review of the Literature基本概念

- 对该领域/专业/方向的问题或研究搜集大量相关资料，通过分析、归纳、整理最新进展、学术见解或建议，做出综合性介绍和阐述。通常包含在引言中。
 - 该领域的研究意义。
 - 该领域的发展脉络。
 - 目前的研究水平。
 - 存在问题及可能的原因。

2.2 Review of the Literature 写作思路

- 研究领域有哪些开创性的工作？是否需要提及这些工作？
- 自这些开创性工作以来取得了哪些进展？
- 最相关的前沿研究有哪些？提及这些研究的最佳顺序？
- 这些最新研究的成就和局限性是什么？
- 这些局限性揭示了哪些研究空白？
- 本文的工作如何填补这些/某一研究空白？

选题 → 调研 → 拓展 → 总结 → 关联 → 区分与结合

2.2 Review of the Literature写作思路

■ 选题

- 选择的主题最好是作者熟悉且对其富有浓厚兴趣的。
- 要具有大量的数据支撑以及文献基础，可以让作者明确主题的关键所在，顺利开题。

■ 调研

- 经过查阅相关资料，搜集整理与主题内容相符合的科学研究。
 - 与论文主题研究相关的学术性文章都属于搜集的范围。
-

2.2 Review of the Literature 写作思路

■ 拓展

- 对研究领域相关文献的信息延伸网络进行分析，仔细阅读相关的、最显著的研究结果和数据最翔实的文章。
- 利用思维导图、图表等形式明确研究的重点内容，从中挑选对写作最有用的信息。

■ 总结

- 对文章进行总结，为写作奠定坚实的信息基础。
 - 根据文章篇幅大小选择两三个关键概念，并对研究相关的重要问题予以详细阐述。
-

2.2 Review of the Literature写作思路

■ 关联

- 将符合研究内容的相关文献的概念予以展现，并探究文献与研究主题之间的关联性。
- 对于之前研究中尚未涉及的议题部分，应该在文献方面进行探索，确保研究有足够的文献依据。

■ 区分与结合

- 对各文献之间的关联性进行有效区分，同时根据个人观点将所有文献结合起来。
 - 注重现有文献和现有研究之间的关系。
-

2.2 Review of the Literature 写作思路

Our work is closely related to the recent works [4]–[6]. In [4] the authors consider the problem of communication over an additive white Gaussian noise (AWGN) channel with the requirement that a wiretapper should not be able to tell

阐述与已有文献的关联，
提供足够的文献依据

In the current paper, we show that the square-root law holds for a broad class of discrete memoryless channels (DMCs).² Furthermore, we provide exact characterizations for the scaling constant of the amount of information with respect to the square root of the blocklength for DMCs as well as AWGN channels, which is not done in [4]–[6].

强调对现有文献的拓展，
展示本研究的创新之处

Because we do not assume a wiretap structure, contrary to [5], in our setting LPD communication is impossible without a secret key. We assume that such a key is available, and are not concerned with its length within the scope of this paper.

We assume that the receiver does know when the transmitter is sending a message. This is a realistic assumption because

One technical difference between [4], [5] and the present work is that the earlier works use total variation distance to measure probability of detection whereas we use *relative entropy*, as [2], [7]. Note that, when the relative entropy is given, the total variation distance can be upper-bounded using

说明与已有文献的区别，
简要概括本研究的假设等，
在给出区别时，可以适当
提供参考文献与可靠结论
进行结合

2.3 Review of the Literature写作技巧

- 如何在众多已有文献中确定代表性成果？
 - 需要考虑已有文献成果与研究主题关联度的大小和文献作者在所涉领域中的知名度。
 - 需要考虑文献来源期刊的知名度及其与你拟投期刊的关联程度等。
 - 和研究主题直接相关的成果、所属领域中重要学者或专家的成果、发表在该领域内高级别期刊上的成果和发表在拟投稿期刊上的成果都可以作为代表性成果来介绍。

2.3 Review of the Literature 写作技巧

- 如何在综述研究现状的基础上指出存在的问题？
 - 对已有研究中存在问题的阐述要与对研究现状的综述相呼应。
 - 找到合适的切入点或角度来说明存在的问题，需要提升研究者自身的知识体系以及对所研究主题的理解程度等。
 - As far as we know , there are no studies on ...
 - Moreover , no attention has been paid to ...
 - These studies have only dealt with the situation in X, whereas our study focuses on the situation in Y.
-

2.3 Review of the Literature写作技巧

- 如何对代表性成果进行恰当的“综述”？
- “综”，简要介绍相关的代表性成果，但不能过于详细：
 - 如果介绍过于详细，篇幅就可能因此变得太长。
 - 造成与结果和讨论中的相应内容的冲突。
 - 简写引言或文献综述中的现状分析，而把那些真正对自己结果分析有用的文献放在结果或讨论中进行分析 and 评述。这样不但会极大丰富自己的分析，同时还会提高自己论文主体（结果、分析及结论）内容在整个论文中所占的比例。

2.3 Review of the Literature写作技巧

- 如何对代表性成果进行恰当的“综述”？
- “述”，要评述已有研究成果，表达自己对已有成果地位或作用等的观点及看法：
 - 表达观点这部分内容实质上就是你向领域内重要学者或专家致敬的地方，难点是“度”的把握。
 - 避免过于负面的评价。

2.3 Review of the Literature 写作技巧

effectively detect the presence of the communication. The information theoretical limits of covert communication have been extended in [14] for a non-coherent fast Rayleigh-fading wireless channel and in [15] for a continuous-time Poisson channel, and in [16] over quantum channels, respectively. In these works, the focus is on the first order asymptotics and the covert constraint is given as an upper bound of KL divergence between the output distributions with and without the presence of communication.

简要介绍已有成果，概述
各项研究得到哪些结果/结
论即可

评述已有成果，总结研究
现状，将上述研究结合起
来加以分析

maximal power constraint that is varying with blocklength n is also provided. Nevertheless, the characterization of the second order asymptotics as a function of the covert constraint δ over AWGN channels is still missing. The difficulty lies in the continuous nature of AWGN channels, where identical independent distributed random coding is no longer applicable in the finite blocklength regime. Instead, continuous distributions with coupled components in n dimensional Euclidean space are involved. As a result, the divergences of these distributions

评述已有成果，总结目前
研究存在的问题/不足/困境
等，以便引出新的研究问
题或思路

2.4 Review of the Literature 评估

- 是否讨论了最前沿的学术动态？
- 是否在众多的文献中找到了代表性文献？
- 是否对代表性文献进行了恰当的综述？
- 是否指出了现有研究成果中存在的问题？
- 是否避免了只提及支持文章假设的文献？
- 是否按逻辑顺序排列文献？
- 是否确保文献在论文中均被引用？

课堂小结：

- 引言的结构：
 - 科学背景：确定研究主题
 - 同行研究：进行批判性文献综述
 - 待解决问题：强调领域空白
 - 本研究的方案：阐明如何研究待解决的问题
 - 论文结构：介绍论文的章节结构。
- 撰写引言的要点和注意事项
- 撰写文献综述的思路和技巧